

Notes on creating translation data to train base models

To build our translation data we used the datasets that had triplets of (MSA - Dialect - English), such datasets were available for the Saudi dialect (Saudial), Palestinian dialect (Maknuune), Syrian dialect (UFAL), but for the remaining two dialects: Moroccan and Egyptian. Triplets of (MSA-Dialect-English) were not available. Instead, we had for the Moroccan dialect (DoDa) pairs of (Dialect - English). We used NLLB to get the MSA translation from the English reference we had in the data, we assumed that MSA to English translation is quite reliable. However the Egyptian dialect didn't have any parallel corpus, in this case we used the MADAR-25 multilingual parallel data to get triplets for the Egyptian dialect as well as all the others, since MADAR dataset is divided into cities instead of countries, we selected the capital of each country as a reference to its entire dialect. Although, after closely examining the entries of MADAR in English, it seemed to us that the translations from MSA to English were not natural and probably made using MT. For example the following sentence

"ممکن آخذ حبوب؟",

Is translated to "May I have some cereals?", a native Arabic speaker, would more likely translate it to "Can I take pills". Similar examples are abundant, but it seems that the dialect <-> MSA pairs are much more reliable.

To build our Gold translation dataset we used only Saudial, Maknuune, UFAL, DoDa (dialect-English pairs only), and Madar25 for Cairo (MSA-Dialect pairs only) this amounted to 376K translation pairs. We had also another bigger set where we used MT on the DoDa dataset pairs to get the MSA translation, and used all Madar25 data, selecting the capital of each country to represent its dialect. Which amounted at the end to 697K translation pairs.

Note that we omitted the Jordanian dialect data, especially the JODA dataset that had (MSA-Dialect) pairs, that we could equate with the Palestinian dialect, because of the geographical and cultural proximity. But we decided not to augment our data with it for the sake of simplicity and avoiding confusions.

Note that the synthetic data that we made was not using MT directly on the dialect, but on English to get the MSA translation, we are aware that opensource MT models perform poorly on dialectal Arabic.

